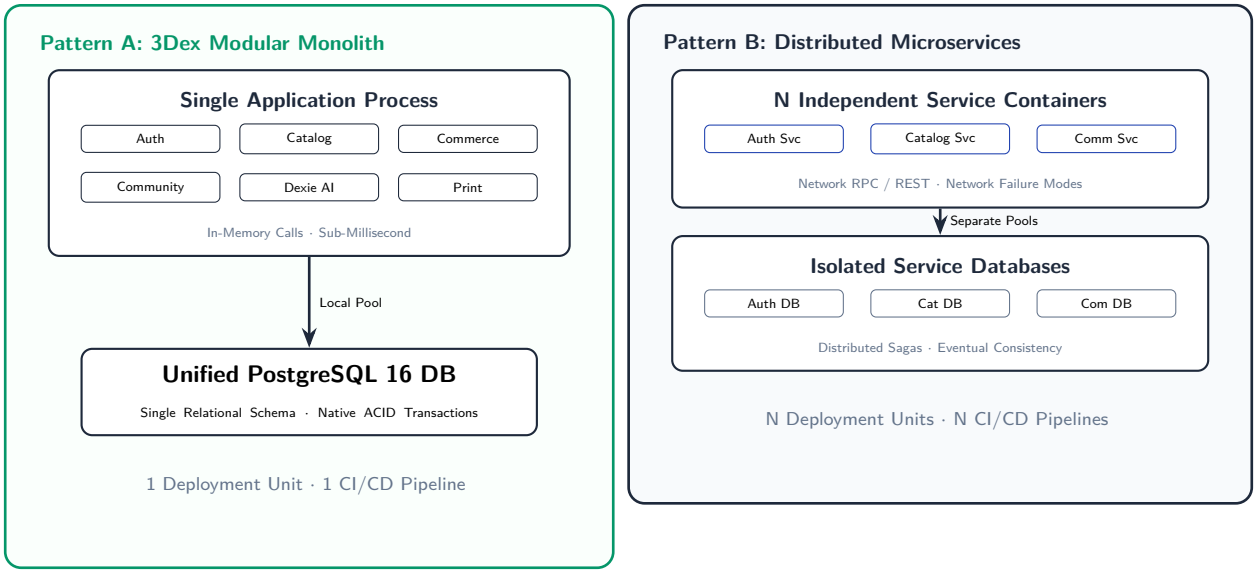
Pattern A: 3Dex Modular Monolith
Single Application Process
Auth
Catalog
Commerce
Community
Dexie AI
Print
In-Memory Calls · Sub-Millisecond
Local Pool
Unified PostgreSQL 16 DB
Single Relational Schema · Native ACID Transactions
1 Deployment Unit · 1 CI/CD Pipeline
Pattern B: Distributed Microservices
N Independent Service Containers
Auth Svc
Catalog Svc
Comm Svc
Network RPC / REST · Network Failure Modes
Separate Pools
Isolated Service Databases
Auth DB
Cat DB
Com DB
Distributed Sagas · Eventual Consistency
N Deployment Units · N CI/CD Pipelines